

Jiayu Xu

Ph.D. Student | Particle Simulation, GPU Computing, and Scientific AI

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Professional Summary

Ph.D. student in Chemical Engineering at the Institute of Process Engineering, Chinese Academy of Sciences, working on multi-GPU parallel computing, discrete element modelling, multiphysics coupling, and AI-assisted experiment-simulation integration. Developer of CUDA/MPI scientific software for large-scale complex-particle systems, with research spanning fundamental algorithms and industrial particulate processes. Participated in industrial research projects with China Coal, Baosteel, and Unilever.

Research Interests

Discrete element method (DEM) and GPU parallel computing
 Non-spherical particles and multiphase-flow simulation
 CFD-DEM coupling and multiscale particle-fluid modelling
 AI-assisted particle identification, tracking, and parameter inference
 Numerical modelling of industrial particulate processes

Education

Ph.D. Student	Institute of Process Engineering, Chinese Academy of Sciences , Chemical Engineering	Beijing, China Sept 2021 – present
	- University of Chinese Academy of Sciences - GPA: 3.70/4.00 - Advisors: Prof. Ji Xu and Prof. Wei Ge - Research on DEM, GPU computing, non-spherical particles, multiphase-flow simulation, and AI-assisted particle analysis	
Bachelor of Engineering	Xiangtan University , Chemical Engineering and Technology	Xiangtan, China Sept 2017 – June 2021
	- School of Chemical Engineering - GPA: 3.73/4.00 - Advisor: Prof. Yefeng Zhou - Research on CFD-DEM coupling, gas-solid fluidized beds, wet granular flows, and cohesive-particle modelling	

Research Experience

Institute of Process Engineering, Chinese Academy of Sciences , Large-Scale DEM and GPU-Accelerated Computing	Beijing, China Sept 2022 – present
- Developed a CUDA/MPI-based multi-GPU DEM framework for concave and convex polyhedral particles - Enabled simulations of 20 million complex particles represented by approximately 12 billion triangular facets - Achieved near-linear weak scaling and over 80 percent strong-scaling efficiency on 16 NVIDIA RTX 4090 GPUs - Applied the framework to rotating drums, silo discharge, fixed beds, and particle packing	

Institute of Process Engineering, Chinese Academy of Sciences, Polyhedral-Particle CFD-DEM and Multiscale Gas-Solid Modelling

Beijing, China
Jan 2025 – present

- Developed a CPU-GPU heterogeneous coupling framework integrating OpenFOAM with a polyhedral-particle DEM solver
- Investigated multiscale gas-solid methods including CG-DPM and MP-PIC
- Targeted fluidized beds, fixed beds, and industrial particle-conveying systems

Institute of Process Engineering, Chinese Academy of Sciences, AI-Assisted Particle Measurement and DEM Parameter Identification

Beijing, China
July 2024 – Feb 2026

- Generated synthetic DEM datasets and trained Mask R-CNN models for irregular-particle image segmentation
- Reconstructed particle trajectories and velocity fields from high-speed experimental images
- Combined Vision Transformers, grid search, and CMA-ES optimization for rapid DEM contact-parameter identification

Selected Projects

Rotary Storage Device for Blast Furnace Slag

May 2024 – May 2026

Industrial research project with Baosteel

- Developed particle-flow models for a high-temperature rotary storage device
- Evaluated feeding mode, storage volume, rotation speed, and lifter geometry
- Supported uniform discharge and equipment optimization

DEM Simulation of Wet Granule Extrusion

June 2022 – Aug 2024

Industrial research project with Unilever

- Developed wet-particle DEM models for basket-granulation extrusion
- Investigated formulation, moisture content, extrusion force, and material flux
- Supported formulation optimization and process scale-up

Quantitative Modelling of Blast Furnace Burden Distribution

Sept 2019 – Nov 2021

Industrial research project with Baosteel

- Modelled blast-furnace charging and burden distribution
- Analysed burden trajectory, surface profile, ore-to-coke ratio, and particle-size distribution
- Supported charging-strategy optimization and coke-ratio reduction

Publications

DEM framework for large-scale concave and convex polyhedral particle systems with multiple GPUs

Jan 2026

Jiayu Xu, Shuai Zhang, Yong Zhang, Haolei Zhang, Duo Zhao, Ji Xu, Wei Ge
[10.1007/s00466-026-02781-y](https://doi.org/10.1007/s00466-026-02781-y) (Computational Mechanics)

Large-scale multiple-GPU-based DEM simulation of polyhedral particle systems

Jan 2026

Jiayu Xu, Shuai Zhang, Yong Zhang, Haolei Zhang, Ji Xu, Wei Ge
[10.1016/j.powtec.2026.122132](https://doi.org/10.1016/j.powtec.2026.122132) (Powder Technology, 472, 122132)

An efficient non-spherical particle tracking strategy based on deep-learning and simulation-experiment integration

Jan 2026

Jiayu Xu, Shuai Zhang, Wei Ge
[10.1016/j.powtec.2025.121681](https://doi.org/10.1016/j.powtec.2025.121681) (Powder Technology, 468, 121681)

Deep-learning-based property parameters identification for DEM simulations

Jan 2026

Jiayu Xu, Shuai Zhang, Haolei Zhang, Ji Xu, Wei Ge
[10.1016/j.ces.2026.124100](https://doi.org/10.1016/j.ces.2026.124100) (Chemical Engineering Science, 332, 124100)

- Multiscale mixing index for granular systems** Jan 2024
Jiayu Xu, Feiguo Chen, Ji Xu, Wei Ge
[10.11949/0438-1157.20240194](#) (CIESC Journal, 75(6), 2214-2221)
- Numerical and experimental investigation of the effect of interstitial liquid viscosity on the collapse of wet granular columns** Jan 2025
 Shuai Zhang, **Jiayu Xu**, Abul Hassan Syed, Leina Hua, Chuan-Yu Wu, Guoping Lian, Wei Ge
[10.1016/j.ces.2024.120725](#) (Chemical Engineering Science, 301, 120725)
- Coupled simulation method of CG-DPM and MP-PIC for gas-solid system** Jan 2025
 Shuai Zhang, **Jiayu Xu**, Leina Hua, Wei Ge
[10.11949/0438-1157.20250599](#) (CIESC Journal, 76(9), 4412-4424)
- Numerical investigation on the fast pyrolysis of large biomass particles in bubbling beds using the coupled particle-scale and coarse-grained CFD-DEM** Jan 2026
 Guanlong Ren, Ji Xu, **Jiayu Xu**, Ying Jia, Wei Ge, Henrik Ström, Jinze Dai, Qingang Xiong
[10.1016/j.cej.2026.177360](#) (Chemical Engineering Journal, 540, 177360)
- Revolution of coarse-grained CFD-DEM technology and its application in fluidized beds: A comprehensive review** Jan 2025
 Guanlong Ren, Ji Xu, **Jiayu Xu**, Yi Ouyang, Henrik Ström, Wei Ge, Haijun Sun, Qingang Xiong
[10.1016/j.partic.2025.08.017](#) (Particuology, 106, 45-69)

Patents and Software

- Invention patent for object-detection methods, devices, electronic equipment, and storage media
- Invention patent for parallel computation of polyhedral DEM systems
- Invention patent for jet-based solid mixing
- polyDEM V1.0-V2.0 - complex-particle DEM simulation software, Registration Nos. 2025SR2431344 and 2026SR0709452
- DEMms V8.0 - GPU-based DEM software, Registration No. 2026SR0445479

Skills

Programming: C/C++, Python, MATLAB, CUDA, and Shell

Parallel Computing: GPU computing, MPI, CPU-GPU heterogeneous computing, and domain decomposition

Numerical Methods: DEM, CFD-DEM, CG-DPM, MP-PIC, and MPS

Artificial Intelligence: Machine learning, deep learning, image segmentation, CMA-ES optimization, and LLM/AI-agent workflows

Software: Linux, Vim, Origin, AutoCAD, 3ds Max, and Ansys Fluent

Languages

Chinese: Native

English: CET-4 564; CET-6 472

Honors and Awards

- Outstanding Graduate of Xiangtan University, 2021
- Zhanchen Enterprise Scholarship, 2020
- First-Class Scholarship of Xiangtan University, 2018, 2019, and 2020
- Third Prize, National College Chemical Engineering Design Competition, 2018